



TY Introduction to Computer Science

Name: _____

Lesson 01

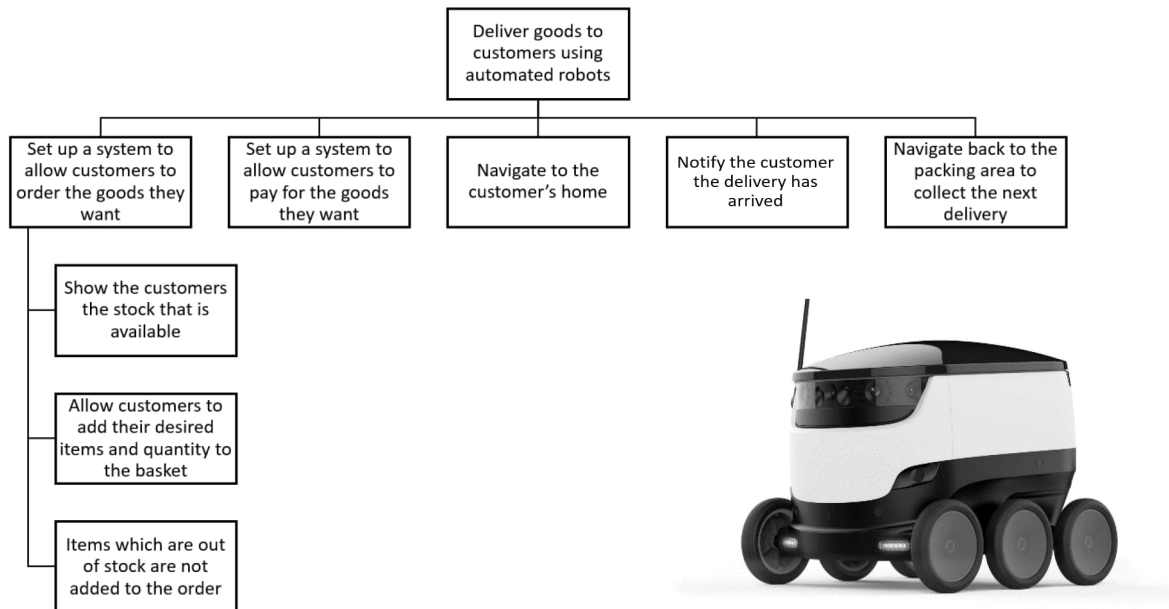
Computational Thinking

Decomposition: Delivery Robot

Name: _____

Class: _____

Decomposition is when you break a task into smaller and smaller tasks. Look at this diagram showing the tasks for creating a delivery robot system.



The main task is to “*Deliver goods to customers using automated robots*”. In order to achieve this, the main task has been broken down into 5 sub-tasks.

What are the sub-tasks that the main task has been broken into?

1.
2.
3.
4.
5.

The sub-task “*Set up a system to allow customers to order the goods they want*” has been broken into 3 further sub-tasks. **What are they?**

1.
2.
3.

You are going to decompose one of the *other* sub-tasks. You can pick any of the other sub-tasks to break it down into smaller sub-tasks.

Which sub-task are you going to decompose?

--

Use the space below to write the sub-tasks for the task you have picked. Remember to write them as sentences to explain the task that needs to be completed, in order to set up a robot delivery system. You do not have to use all the spaces allocated below.



Pattern Recognition: Finding Order

Name:

Class:

Part of **pattern recognition** is to put instructions into a logical order. Below is a recipe for making a really delicious soup. Read all the instructions first and then number them from 1 to 9 to put them in the correct order to make the soup.

Tomato Soup

Cooking Instructions	Order (1 to 9)
Simmer the chunky soup for 15 minutes to cook all the vegetables, before adding some chopped fresh basil leaves, right at the end.	
Add the sliced onions to the warmed oil and cook for 1 – 2 minutes. And garlic and cook for a further 2 minutes, stirring regularly.	
Once you have sliced the onion and garlic, pour 1½ tablespoons of olive oil into a large saucepan and gently warm.	
Serve in bowls with chunks of fresh bread.	
Add 2 chopped carrots and 2 medium chopped potatoes to the onion and garlic mixture and cook for a further 2 minutes string well.	
Peel and finely slice 2 onions and 3 cloves of garlic.	
Remove from heat once the chunky soup is cooked and leave to cool. Once cool, transfer to a blender. Blend the soup until it is smooth. If you try to blend it whilst it is too hot the soup can explode from the blender and may scald you.	
Add 2 tins of chopped tomatoes to the pan to join the onions, garlic, carrots and potatoes. Add 1 litre of vegetable stock and 1 tablespoon of maple syrup and bring the mixture to the boil.	
Return the smooth soup to the pan and warm through.	

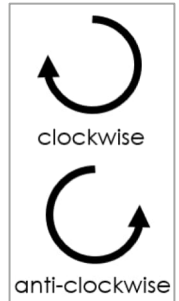


Pattern Recognition: Spot the Correct Pattern

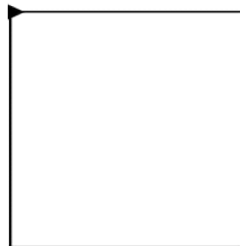
Name:

Class:

Look at each shape. Tick which of the sets of instructions is correct to draw that shape. The small triangle shows the starting point and direction the drawing tool is facing. Only one set of instructions is correct for each shape.



Shape 1



Pen down
Forward 50
Turn 90° clockwise
Forward 50
Turn 90° clockwise
Forward 50
Turn 90° clockwise



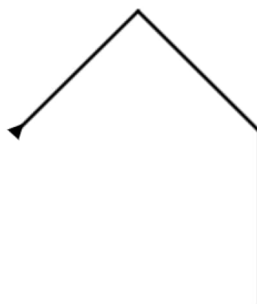
Pen down
Repeat 4
Forward 50
Turn 90° clockwise
Pen up



Pen down
Repeat 4
Forward 50
Turn 90° clockwise



Shape 2



Pen down
Forward 50
Turn 90° clockwise
Forward 50
Turn 45° clockwise
Forward 50



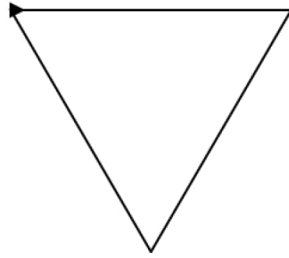
Pen down
Repeat 3
Forward 50
Turn 90° clockwise
Forward 50



Pen down
Forward 50
Turn 90° anti-clockwise
Forward 50
Turn 45° clockwise
Forward 50

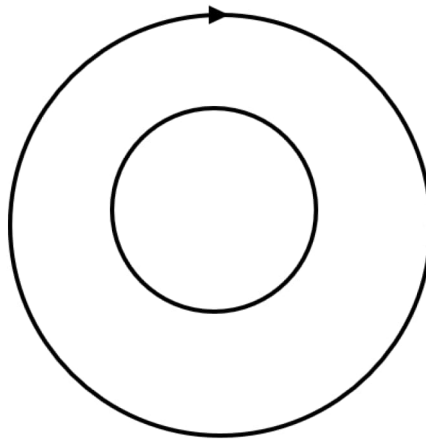


Shape 3



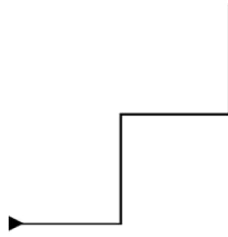
Pen down Forward 50 Turn 60° clockwise Forward 50 Turn 60° clockwise Forward 50 Turn 60° clockwise 	Pen down Repeat 3 Forward 50 Turn 120° clockwise 	Pen down Repeat 4 Forward 50 Turn 60° clockwise 
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Shape 4



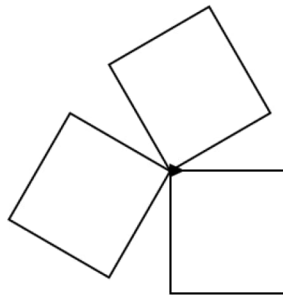
Pen down Repeat 360 Forward 2 Turn 1° clockwise Turn 90° clockwise Move 50 Repeat 360 Forward 1 Turn 1° clockwise 	Pen down Repeat 360 Forward 1 Turn 1° clockwise Pen up Turn 90° clockwise Move 50 Turn 90° anti-clockwise Pen down Repeat 360 Forward 1 Turn 1° clockwise 	Pen down Repeat 360 Forward 2 Turn 1° clockwise Pen up Turn 90° clockwise Move 50 Turn 90° anti-clockwise Pen down Repeat 360 Forward 1 Turn 1° clockwise 
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Shape 5



Pen down Forward 25 Turn 90° clockwise Forward 25 Turn 90° clockwise Forward 25 Turn 90° clockwise Forward 25 Turn 90° clockwise □	Pen down Repeat 4 Forward 25 Turn 90° clockwise □	Pen down Repeat 2 Forward 25 Turn 90° anti-clockwise Forward 25 Turn 90° clockwise □
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Shape 6



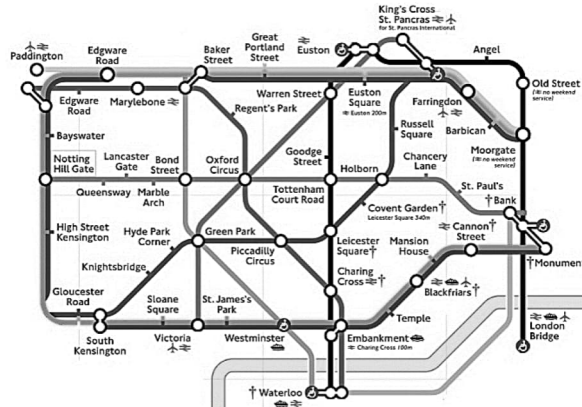
Pen down Repeat 3 Forward 25 Turn 90° clockwise Turn 120° clockwise Repeat 3 Forward 25 Turn 90° clockwise Turn 120° clockwise Repeat 3 Forward 25 Turn 90° clockwise Turn 120° clockwise Repeat 3 Forward 25 Turn 90° clockwise Turn 120° clockwise □	Pen down Repeat 3 Repeat 4 Forward 25 Turn 90° clockwise Turn 120° clockwise □	Pen down Forward 25 Turn 90° clockwise Forward 25 Turn 90° clockwise Turn 120° clockwise Forward 25 Turn 90° clockwise Forward 25 Turn 90° clockwise Forward 25 Turn 120° clockwise Forward 25 Turn 90° clockwise Forward 25 Turn 90° clockwise Turn 120° clockwise Forward 25 Turn 90° clockwise Forward 25 Turn 90° clockwise Forward 25 Turn 120° clockwise □
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Abstraction: London Underground Map

Name:

Class:

Abstraction is where you focus on the important details in a problem only. Take the London underground map. Here is the official London underground map.



The official London underground map is a simplified version of what the underground actually looks like.

If the map was more similar to reality, it would look as follows:



What do people use the official London underground map for?

What details are missing from the official London underground map?

Why do you think the official London underground map has been simplified?

Abstraction: Gathering Important Facts

Name:

Class:

Read the following statement and then answer the questions below:

"Well, I was walking down the road, not this road, you know the road next to the gym. My brother likes to go to that gym, he's really into his keep fit, and watches what he eats too. He wants to be a personal trainer so it seems like he is always in the gym these days. Well, I was going down there the other night, I think it was Tuesday, or maybe it was Thursday. I know it wasn't Wednesday because I was out with my mate, Emma and we went to the cinema to watch the new Spiderman film. It was really good, I love an action movie. Anyway, I was walking down the road, you know, next to the gym. And that's when I saw it. It was just tied to the lamppost. It was so cold, it was trembling poor thing. I looked around but it looked like nobody owned it. How could anybody leave an animal tied up like that, with no food or water? I untied it and it was so excited it was wagging its tail all the way back to my house. And when we got home it was barking and running around in circles in excitement."

What did they find?

How do you know what they found?

How many siblings do they have? And how do you know this is correct?

Write three more questions you could ask to find out more about the IMPORTANT facts about the story.

1.

2.

3.

Lesson 02

Algorithms

Algorithms: Writing Instructions

Names:

Class:

Work in pairs and design your own unique secret handshake involving between 5 and 10 different actions.



Write instructions using **full sentences** in the space below, to describe how to perform those actions.

Once you have written your instructions, test them out on other students by only reading out your instructions to see if they can replicate your secret handshake accurately. **You are not allowed to give them any other instructions and you are not allowed to demonstrate your secret handshake or any individual actions yourselves.**

Then answer these questions:

What instructions did they find easy to follow?

Did they get anything wrong?

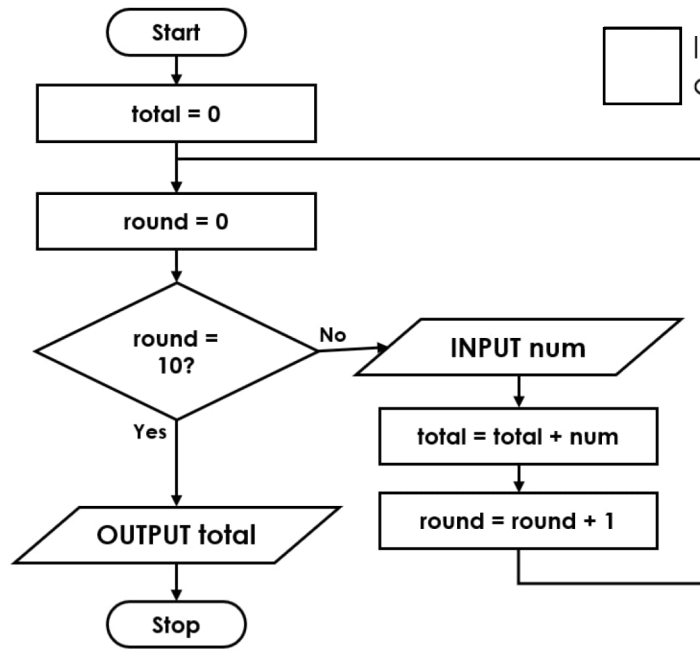
How could you have improved your instructions?

Algorithms: Flow Diagrams

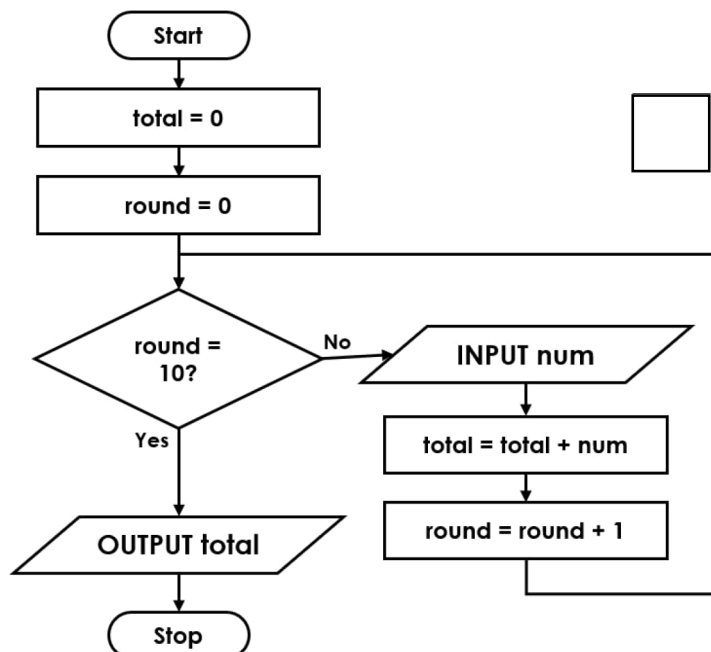
Name:

Class:

Task 1: Here are two flow diagrams. Tick which one is correct to ask a user to enter 10 numbers, add them together and show the total.

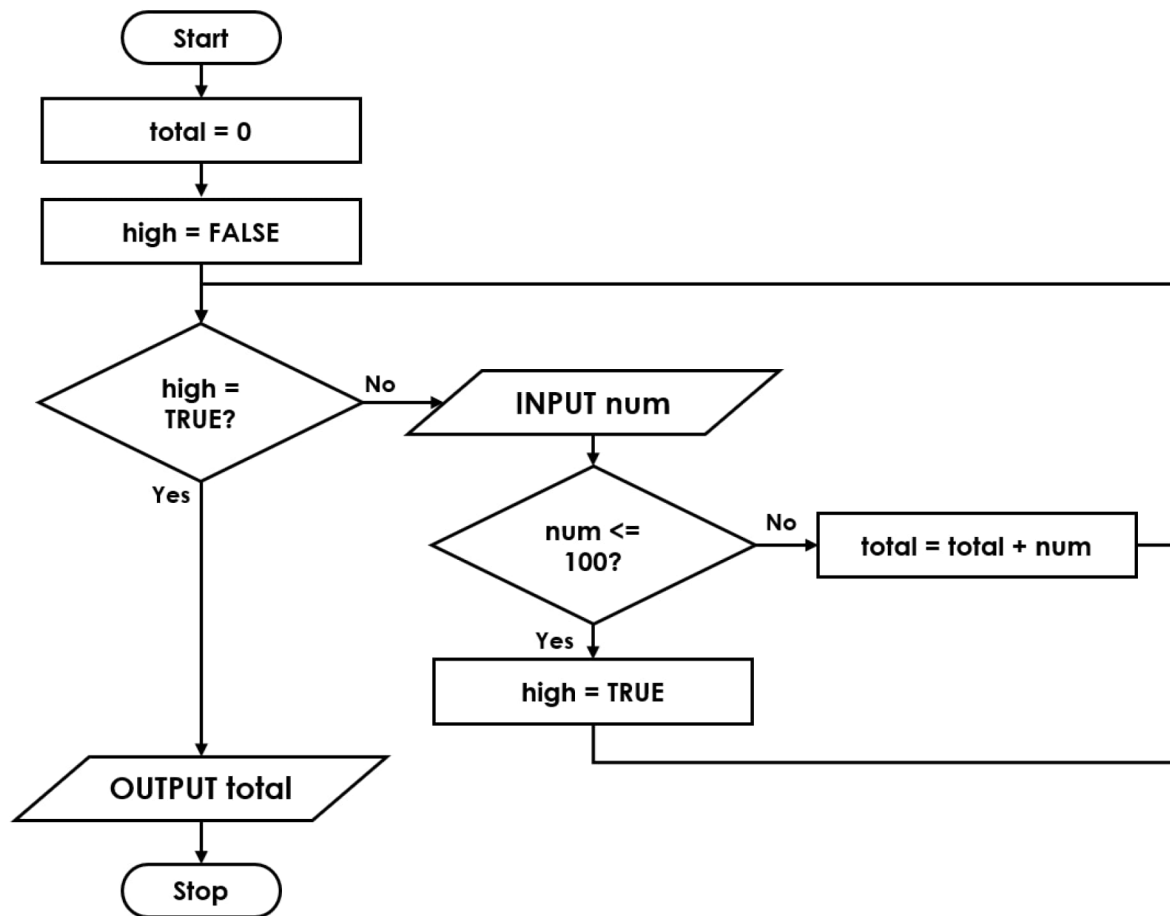


☐ I think this flow diagram is correct

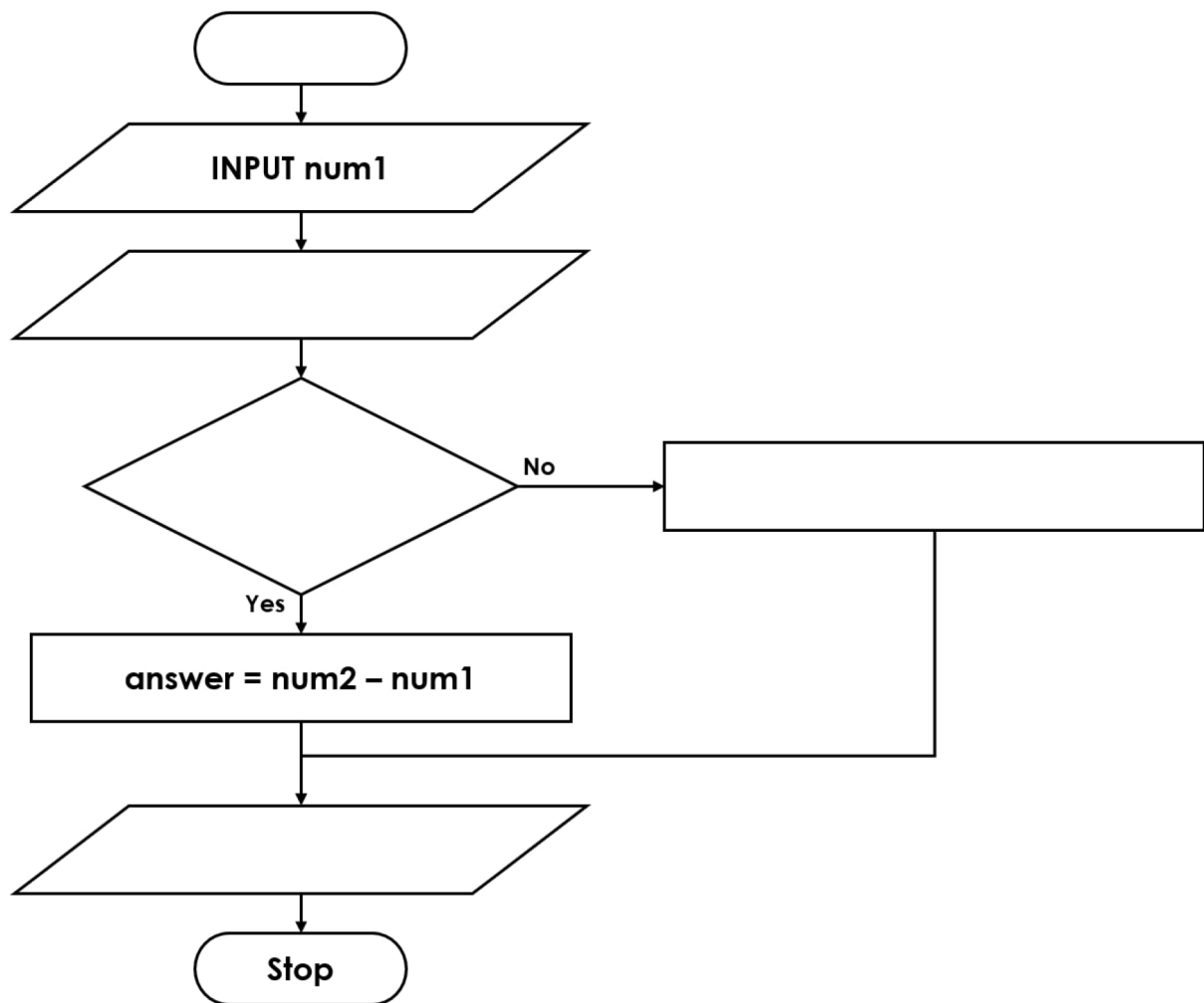


☐ I think this flow diagram is correct

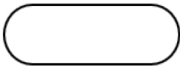
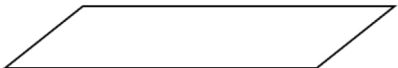
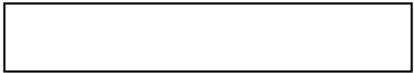
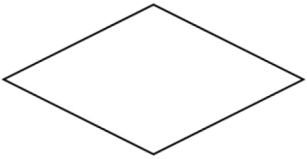
Task 2: Use the space below to describe what you think this flow diagram will do.



Task 3: Fill in the gaps for the following flow diagram. The flow diagram should ask the user to enter two numbers. Subtract the smallest number from the larger one and display the answer.



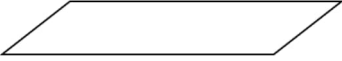
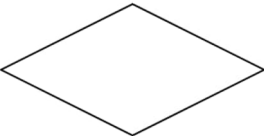
Task 4: What do each of these symbols represent in a flow diagram?

Symbol	What does it represent?
	
	
	
	

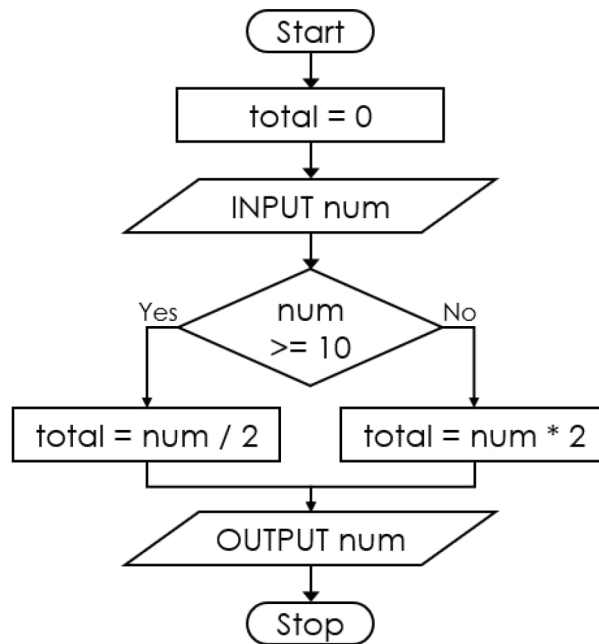
Flow diagrams

Name: _____

Class: _____

Symbol	What does it represent?
	
	
	
	

Read the following flow diagram and fill in the truth table for each of the different inputs listed below.



Input	Output
4	
14	
23	
9	
10	

Lesson 03

Online Safety

Social media account

Here is a fake social media account. Look through the available information and see what you can discover about this person.

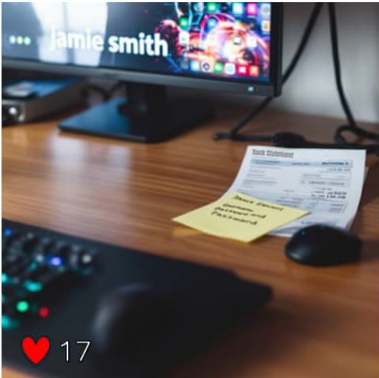
Front page


InstaSnap



@CoolKid2025
Love football 🏈 | Year 11 | 📍 London | DM for my number!


❤️ 23
First day of Year 11! So excited! 🏠


❤️ 17
Check out my new gaming setup! ...


❤️ 38
Holiday vibes! Tropical paradise ...

Their profile


InstaSnap

Username: @CoolKid2025
Location: London, UK

Bio: Love football 🏈 | Year 11 |
📍 London | DM for my number!

Hobbies:

- Playing football for St. Francis School.
- Gaming (especially Minecraft and FIFA).
- Watching YouTube videos about gaming tips and tricks.
- Hanging out with friends at the park.

Post 1



@CoolKid2025 2nd September

First day of Year 11! So excited! 🏠

Likes: 23



@Friend1: Good luck in Year 11!



@UnknownUser: Nice house! Where is it?



@CoolKid2025: thanks, I don't give out my address online.

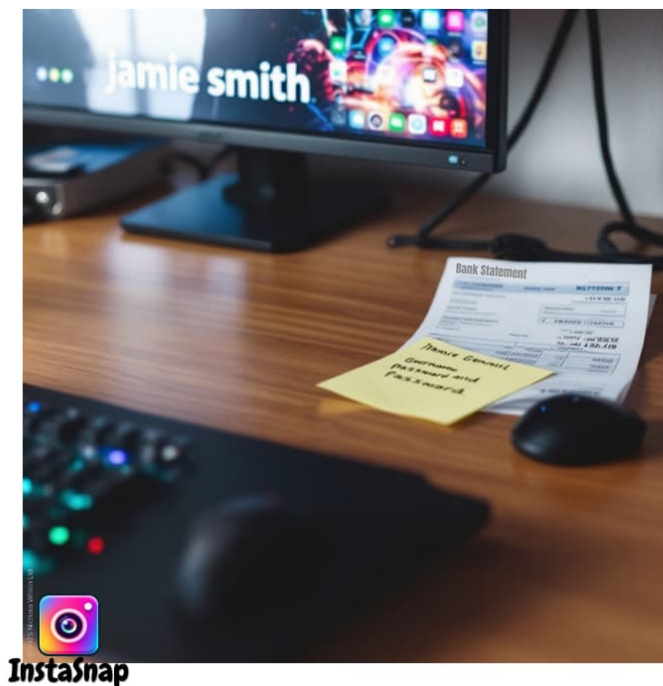


@UnknownUser: I don't want the address, was just wondering the area



@CoolKid2025: oh, it's just round the corner from my school.

Post 2



@CoolKid2025 12th April

Check out my new gaming setup! 🎮 DM me if you want to play!

Likes: 17



@Friend2: Wow, that setup looks awesome! What games do you play?



@Friend3: Can't wait to see it in action! Let's play this weekend!



@UnknownUser2: "Cool setup! What's your gaming username?"



@CoolKid2025: jsmith29462



@UnknownUser3: "Do you stream your games? Would love to follow!"



@CoolKid2025: yes, check out my YouTube channel jsmith_coolkid2025

Post 3



@CoolKid2025 15th July

Holiday vibes! Tropical paradise 🌴 Back home on 29th, but for now, it's all about the beach life!

Likes: 38



@Friend1: Wow, the Maldives looks amazing! Hope you're having the best time! 🐦



@Friend3: Super jealous! Bring back some sunshine with you! ☀️ Also Happy Birthday for Thursday.



@CoolKid2025: Thanks Arjun



@UnknownUser2: "Looks beautiful! Are you travelling alone or with family?"



@CoolKid2025: I'm with my fam.



@Friend2: "Don't forget to take loads of pictures! Can't wait to see them when you're back!"

Becoming a digital citizen

Name:

Class:

1: Digital footprint

Here is a list of actions related to online behaviour. For each action, decide whether it creates a positive or negative digital footprint and tick the appropriate column.

Action	Positive	Negative
Writing a supportive comment on someone's blog post.		
Sharing a funny meme that includes someone's personal photo without their permission.		
Complaining about a teacher or classmate publicly on social media.		
Posting a video tutorial you created to help others with homework.		
Using someone else's artwork in your own post without crediting them.		
Reporting an inappropriate or harmful post to a website administrator.		
Sharing your gaming username and password with a friend.		
Posting an inspiring story about overcoming a challenge.		

2: Personal privacy

Write down three things you can do to protect your privacy when online.

Spotting fake news

Name:

Class:

1: Fake news

Create your own fake news headline that contains at least one clue that it isn't real.

Consider using clickbait words to grab attention, spelling or grammar mistakes, out of date news, an unreliable source, emotional manipulation or make it sound too good or bad to believe.

Date of article:

Source of article:

Headline:

Pass your headline to a partner and ask them to identify the specific clues that give away it is a fake.

2: Identifying reliable vs. unreliable sources

Write 3 tips to help someone decide if a website is reliable.

Each tip should be short (1-2 sentences) and easy to understand.

Use examples to make your tips clearer, e.g., “A trusted website might ...”

Safe searching online

Name:

Class:

1: Search engines

You are going to find out the answers to the questions below. You will do this by working in pairs. One pupil searches and the other observes how they search and makes a note of the exact keywords that were typed in.

Question	Answer	Keywords Used
1. What is the tallest mountain in South America and how high is it?		
2. Which country has the most official languages and how many does it have?		
3. In what year was the first email sent and who sent it?		
4. Which animal is known as the "ship of the desert" and why is it called that?		
5. What is the smallest country in the world by land area and where is it located?		

2: Search engines

Work in pairs to think of a question or topic to search for. For example: "What are the benefits of renewable energy?" or "Who invented the internet?" Make sure it's something you're curious about and that will give detailed results.

Use Google for your first search and use Bing for your second search.

Type the exact same query into both search engines and look at the first 5 results from each search engine and then answer the questions below.

Question	Answer
1. Are the results the same or different?	
2. Do the top results come from well-known websites or lesser-known ones?	
3. Are there any advertisements at the top of the page?	
4. How easy is it to find the answer to your question?	

3: Search Engines

Write 2–3 sentences explaining why search engines might not always show the best or fairest results.

Lesson 04

AI in Education

Using AI Wisely in School

Student name: _____ Date: _____ Pair/Group: _____

Lesson goals:

- Understand what AI/LLMs are in plain language.
- Identify advantages and problems of using AI in school.
- Create effective prompts.
- Check AI answers for errors and hallucinations.
- Practice verifying information accuracy.

Part A — AI / LLM: What is it?

1. In your own words, what is AI?

2. Large Language Models (LLMs): underline or tick all that apply

- ☐ Predict likely next words based on patterns in data
- ☐ Always know the truth
- ☐ Can generate text, summaries, and explanations
- ☐ Can make mistakes confidently (hallucinate)
- ☐ Replace learning

3. Where have you seen AI used (school or everyday)?

Part B — Advantages and Problems (Pairs)

Sort each item into “Advantage” or “Problem” and add one example of your own.

- Fast idea generation: Advantage or Problem

because:

- Personalized explanations: Advantage or Problem

because:

- Can be wrong or biased: Advantage or Problem

because:

- Privacy risks: Advantage or Problem

because:

- Drafting structure for essays/reports: Advantage or Problem

because:

- Over-reliance/shortcut temptation: Advantage or Problem

because:

Part C — Prompt Generation

Task: Write two prompts on the same topic, one weak and one strong. Then improve the weak one.

Topic: Choose one from the list at the end of the worksheet

1. Weak prompt (intentionally vague):

2. Strong prompt (clear goal, audience, format, constraints):
Include: audience level, length, bullet points vs paragraph, 2 examples, one common misconception.

3. Improve the weak prompt by adding at least 3 constraints:

4. Predict: What error might the AI still make with your strong prompt?

Part D — Error Checking / Hallucination

Identify red flags and plan verification steps.

1. Hallucination red flags (tick those you've seen/heard of):

- ☐ Overconfident tone with no sources
- ☐ Fake citations or broken links
- ☐ Specific numbers with no references
- ☐ Misattributed quotes
- ☐ Mixing up similar terms (e.g., correlation vs causation)

2. For each red flag you ticked, write one way to check it:

3. Write a “verification prompt” you would send back to the AI to self-check its answer:
Example starter: “List your sources with working links and explain how each supports claim X.”

- Your verification prompt:

Part E — Practical Exercise: Information Accuracy (small groups)

Goal: Test AI accuracy on a factual topic and document your verification.

1. Choose a focused question (specific, checkable):
Examples: “What were the main terms of the 1919 Treaty of Versailles?” or “What are the health impacts of vaping among teens according to 2022–2024 studies?”
 - Your question:

2. Your initial prompt to the AI (clear, with constraints):
Include: audience level, ask for 3–5 bullet points, require sources with working links, and dates.

3. Record the AI’s key claims (3–5 bullets) and the sources it gave:

- Claim 1: _____ Source: _____
- Claim 2: _____ Source: _____
- Claim 3: _____ Source: _____

4. Verify each claim using at least two of:

- The source link provided
- A trusted reference site (encyclopaedia, government, academic)
- Your textbook or class notes
- A library database (if available)

Evidence log:

- Claim 1: Verified? Yes / No — Evidence source(s): _____
- Claim 2: Verified? Yes / No — Evidence source(s): _____
- Claim 3: Verified? Yes / No — Evidence source(s): _____

5. If something didn't verify:

- What was wrong or missing?

- Follow-up prompt you sent to correct it:

6. Final summary (3–4 sentences): What is accurate, what is uncertain, and what you'd cite.

Part F — Responsible Use Reflection (5 minutes)

1. One classroom scenario where AI use is appropriate and how you would cite it:

2. One scenario where AI should not be used, and why:

3. Your personal rule for using AI this term (one sentence):

Mini checklist — Accuracy and Ethics

- I requested sources with dates.
- I clicked links and verified claims.
- I noted any uncertainties.
- I cited AI assistance if used.
- I protected privacy and avoided sensitive data.

Prompt Topic List

Here are varied, subject-neutral topic prompts students can choose from for Part C (Prompt Generation). They're broad enough for any class but specific enough to craft strong prompts.

Core academic

- Climate change: drivers, impacts, and mitigation examples
- Photosynthesis vs. cellular respiration (differences and misconceptions)
- Causes of World War I vs World War II (compare/contrast)
- The Treaty of Versailles: key terms and consequences
- Pythagoras' theorem: everyday applications beyond triangles on paper
- Plate tectonics: causes of earthquakes and volcanoes
- The water cycle: human impacts on each stage
- The periodic table: trends (groups vs periods) and why they matter
- Probability in real life: risk vs uncertainty (common misconceptions)
- Media bias: indicators and how to cross-check

Digital literacy and computing

- Password security: what makes a password strong and why
- Phishing: how to spot it and what to do
- Digital footprints: how data brokers collect and use data
- AI in search engines: how results are ranked and risks of "answer engines"

- Algorithmic recommendations: benefits and echo chamber risks
- Deepfakes: how to detect manipulated media
- Copyright and fair use: what students need to know for school projects
- Cyberbullying: reporting pathways and bystander strategies

Health and wellbeing

- Teen sleep: effects on learning and mood (recent evidence)
- Nutrition labels: how to read them and common marketing tricks
- Vaping vs smoking: comparing health risks (with sources)
- Screen time: impact on attention and sleep (what evidence actually says)
- First aid basics: recovery position and when to use it
- Hydration and performance: myths vs facts

Civics and ethics

- Freedom of speech vs harmful misinformation: where lines are drawn
- Data privacy laws basics (GDPR/age-appropriate design): what applies to students
- Facial recognition in public spaces: benefits and ethical concerns
- Digital citizenship: rights and responsibilities online
- Sustainable consumption: fast fashion impacts and alternatives

Arts and culture

- Music streaming algorithms: how they shape discovery
- Copyright in fan art and remixes: what is allowed?
- Cultural appropriation vs appreciation: how to tell the difference
- Public domain: what it is and how creators can use it
- Museum curation: how exhibits shape historical narratives

Geography and environment

- Urban heat islands: causes and simple mitigation tactics
- Coastal erosion: human and natural drivers with local examples

- Plastic pollution: microplastics and health/ecosystem impacts
- Renewable energy types: pros/cons and situational fit
- Sustainable transport: comparing emissions and accessibility

Study skills and exam literacy

- Effective note-taking methods: Cornell vs outline vs map
- Spaced repetition vs cramming: what research shows
- Growth mindset vs fixed mindset: classroom applications
- Academic integrity in the age of AI: appropriate vs inappropriate use

Finance and life skills

- Budgeting basics for teens: needs vs wants
- Credit and interest: how debt grows over time
- Consumer scams: red flags and reporting
- Renting vs buying: pros/cons in different contexts

Science and society

- Vaccination: herd immunity explained simply
- GM foods: common myths vs scientific consensus
- Nuclear energy: risks, benefits, and waste management
- Ethical AI: bias, transparency, and accountability in schools

Ready-to-use prompt stem students can apply to any topic above

- “Explain [chosen topic] for a 15-year-old in ≤120 words. Include 1 analogy, 2 everyday examples, and 1 common misconception. Provide 2 sources with titles, authors, dates, and working links.”
- “Compare [A] and [B] in 4 bullets: definition, two key differences, one common misconception, one-sentence takeaway. Include 1 credible source with a working link.”

- “Give 3–5 key facts about [topic] from 2020–2025 sources. For each fact, include the source title, organization, date, and a working link. Note any uncertainties.”

Lesson 05

Cyber Security

Cyber threats

Name:

Class:

1: Malware

You are a malware detective tasked with solving four cybercrime cases. Each case involves a different type of malware. Read the clues, identify the type of malware and explain your reasoning. Then, suggest a solution or precaution to help prevent the attack from happening again.

Case 1: The School Network Shutdown

The school's computers suddenly stopped working after a student plugged in a USB stick they found. Files on the system started duplicating themselves and strange error messages appeared.

1. What type of malware is responsible for this attack?

2. Explain how this malware works and why it caused the school network to shut down.

3. What steps could the school take to prevent this from happening again?

Case 2: The Locked Files Mystery

A local business finds that all their important files are locked and a message on their screen demands £500 to unlock them. They cannot access their data and the message warns that the files will be deleted if they do not pay.

1. What type of malware is responsible for this attack?

2. Explain why this malware is effective and what damage it can cause.

3. What advice would you give the business to recover their files and avoid this in the future?

Case 3: The Fake Game Trap

A student downloaded a free game from an unverified website. After installing it, their computer started running slowly and their saved passwords for social media accounts were stolen.

1. What type of malware is responsible for this attack?

2. Explain how this malware tricked the student and what damage it caused.

3. What should the student do to protect their computer and personal information?

Case 4: The Sneaky Spy

A teacher noticed that their computer was acting strangely. They discovered that someone had been secretly monitoring their activity, including the websites they visited and their keystrokes.

1. What type of malware is responsible for this attack?

2. Explain how this malware operates and why it is dangerous.

3. What tools or actions can the teacher use to detect and remove this malware?

2: Social engineering

You receive a phone call from someone claiming to be from your school's IT department. They say your account has been hacked and ask for your password to fix it immediately.

1. Write down the action you would take instead of giving them your password.

2. Explain why you should never share your password over the phone.

3: Hacking

Complete this table to explain what motivates each type of hacker.

Type of hacker	What motivates them?
White hat	

Grey hat	
Black hat	

Complete this table to explain the impact it has on the owners of the account they hack.

Type of hacker	What are the consequences to the owner?
White hat	
Grey hat	
Black hat	

Reducing Risks

Name:

Class:

Describe how you can reduce the risks of the following dangers.

Danger	How can you reduce the risk?
Staff accessing data they should have no right to see	
Staff losing a USB with confidential private data	
A virus that deletes files on your computer system	
Cybercriminals finding a way into your network as you are using an old version of the operating system	
Cybercriminals gaining entry to your computer network as some of your staff are using "password" as their password	

Dark web and cybercrime

Name:

Class:

1: The dark web

Name two differences between the surface web and the dark web.

- 1) _____

- 2) _____

2: Stolen data

Complete each sentence by filling in the missing word from the word bank:

passwords	marketplaces	credit card
dark web	criminals	personal

Cybercriminals often steal _____ information from people and businesses.

Stolen data, such as _____ details or _____, can be sold online.

Illegal _____ are hidden websites on the _____ where
_____ buy and sell stolen data.

3: Risks

If you had the power to make one new law or rule to help reduce cybercrime on the dark web, what would it be?

- Explain why you chose this rule.
- How do you think it would help protect people?
- Are there any possible downsides or challenges with your idea?

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.